# Can a Simple Neural Network Improve Cohort Fertility Forecasts?

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**Abstract**

Forecasting completed cohort fertility (CCF) helps resolve ambiguities in total fertility rate (TFR) trends, yet conventional cohort forecasting methods perform little better than a freeze-rate baseline (Bohk-Ewald et al., 2018). Inspired by deep learning advances in mortality forecasting, this paper tests whether a Multi-Layer Perceptron (MLP), named CCF-Net, can improve cohort fertility projections. Using data from the Human Fertility Database and evaluated on a well-established cohort fertility ranking system, CCF-Net outperforms or matches top demographic benchmarks. Across the historical jump-off years tested, its higher forecasts are closer to true outcomes by better accounting for rises in cohort fertility in the late-1990s and early-2000s. Conversely, when trained on data up to 2019, CCF-Net predicts substantially and perhaps unrealistically lower future CCF than traditional models, which may call into question whether its good performance will continue going forward.

## 1 Introduction

Fertility assumptions underpin long-run projections of pension solvency, healthcare expenditure, education provision, and labor markets, yet fertility remains notoriously difficult to forecast. Some of the difficulty is that a decline in period fertility, the total fertility rate (TFR), is ambiguous: it may mean that women are having fewer children, or only that they are having them later. Projecting completed cohort fertility (CCF) resolves this ambiguity since it counts the children ever born to a cohort of women regardless of when they arrive (Myrskylä et al., 2013). It is often the quantity that demographic forecasting efforts target.

Yet the available methods for forecasting CCF are surprisingly weak. Only a handful outperform the naive strategy of freezing age-specific fertility rates at their last observed value. Those that do are two extrapolation-based methods (de Beer, 1985, 1989; Myrskylä et al., 2013) and two Bayesian methods (Schmertmann et al., 2014; Ševčíková et al., 2016). More surprising still, the extrapolation approaches match or beat the Bayesian approaches (Bohk-Ewald et al., 2018), despite key features of Bayesian approaches that in principle

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should improve forecast accuracy (partial pooling, informative priors, etc.). Part of the remaining difficulty stems from fertility's non-linearity. While mortality improvements proceed relatively linearly (Lee & Carter, 1992; Oeppen & Vaupel, 2002), fertility patterns feature more sharp shifts, including "rises, falls, and reversals" (Myrskylä et al., 2013). Classic demographic forecasting approaches, often constrained by strong functional form assumptions, can struggle to capture these non-linear trajectories.

Deep neural networks are a promising and as yet untested alternative approach to CCF forecasting that have the potential to better capture these complex fertility trends. In the adjacent domain of mortality forecasting, neural networks have repeatedly outperformed extensions of the classic Lee-Carter framework that have accumulated over three decades of incremental refinement (Zheng et al., 2025). Their advantage appears to come from flexibility; they map the mortality surface more closely, capturing non-linear trends and cohort patterns that classical models miss (Richman & Wüthrich, 2021). Deep learning approaches are increasingly being adopted in actuarial research and practice.

Fertility forecasting, by contrast, is not a central actuarial task, which may explain why neural network methods are developing there far more slowly. However, recent work in demography is beginning to test the capacity of neural networks for fertility projection, finding that a recurrent neural network achieves lower out-of-sample point-forecast errors than a naive drift baseline and the UN's Bayesian hierarchical model for forecasting the TFR (Ciganda et al., 2026). Yet, no study has benchmarked any deep learning methods against classic methods for *cohort* fertility forecasting.

While neural networks are performing well in the realm of mortality forecasting, it is an open question whether they can learn the less stable fluctuations of fertility from substantially less data. I draw on data from the Human Fertility Database (HFD, 2026), which provides age-specific births, exposures, and fertility rates for over 30 countries with high-quality vital statistics reporting. The HFD is the fertility equivalent of the Human Mortality Database (HMD), the data source at the center of much of the deep learning work on mortality forecasting. A major difference between these sources is the total size of the age-specific data; the HFD is smaller (fewer ages and single sex). Because the performance of neural networks is often linked to having enough data from which to learn patterns (Sun et al., 2017), it is unclear whether the success that deep learning has had for mortality forecasting will hold for fertility forecasting, at least if only using HFD data.

Here, I compare performance of the simplest deep learning architecture, a Multi-Layer Perceptron (MLP), with that of the top-performing methods for forecasting CCF as established by the seminal Bohk-Ewald et al. (2018) paper. I call the chosen neural network the Completed Cohort Fertility Neural Network or "CCF-Net," and ask whether its flexibility and its ability to pool information across countries translate into better cohort fertility forecasts. The evaluation approach taken here ensures that metrics were not cherry picked to demonstrate superior performance, but instead match those used in the 2018 reference paper. I use code and evaluation criteria from Bohk-Ewald et al. (2018) to re-estimate the top-performing cohort fertility forecasting models before adding CCF-Net to the model suite. Because the benchmark models and accuracy measures are inherited rather than chosen, CCF-Net's performance is assessed against approaches and criteria the field has already established as important. The primary goal of the paper is to place CCF-Net on demography's informal fertility forecasting "leaderboard" and to qualitatively examine CCF-Net's

historical and future forecasts.

This paper makes three key contributions. First, I find that CCF-Net outperforms or matches all benchmark methods across the criteria established by Bohk-Ewald et al. (2018) and its advantage grows over forecast horizon. Second, I qualitatively evaluate CCF-Net's forecasts finding that on average, its forecasts are higher than all other methods, which in most evaluation periods considered is closer to observed cohort fertility. CCF-Net seems to better capture rises in completed fertility among cohorts 1975-1985, and does especially well in post-socialist, Eastern European countries. Third, I show that while CCF-Net produces slightly more accurate forecasts on historical data, its error is less stable across jump-off years and its forecasts of future, unobserved cohort fertility appear surprisingly low.

# 2 Methods

## 2.1 Data

All models are estimated on age-specific fertility rates (ASFR) from the Human Fertility Database (HFD, 2026). Two Lexis shapes of the same underlying data are used:

- Period rates on Lexis squares (asfr/asfrRR.txt): fertility by single year of calendar time and completed age at the end of the year (ACY). This is the primary input for six of the seven methods and is the primary data type used in Bohk-Ewald et al. (2018) as well.
- Cohort rates on vertical parallelograms (asfr/asfrVV.txt): fertility by single year of age reached during the year (ARDY) and birth cohort. This is the native cohort format and is used by the Schmertmann et al. model, which is specified on an age × cohort surface.

The data spans calendar years 1891–2024; all cohort analyses are restricted to ages 15–40. While childbearing past 40 is increasingly common, I retain this restriction to maintain alignment with the reference paper. Period rates are converted to cohort rates by the identity "cohort = calendar year − age", so the rate of cohort $c$ at age $a$ is the period rate observed in year $c + a$. This conversion method is in line with the prior literature (Bohk-Ewald et al., 2018). Countries are only included if they observe every age from 15 to 40 and are gap-free from their first observed year up to the jump off year.

## 2.2 Forecasting design

Every method is run under a rolling origin design. All jump-off years (JOY) between 1985 and 2013 are tested. At each jump-off year a method sees only data observed up to and including that year, and forecasts calendar years JOY+1 … JOY+30. Period forecasts are then converted to cohort forecasts by the cohort identity above.

Because every method forecasts from a fixed jump-off year, cohorts enter the evaluation at different stages of their childbearing. I refer to a cohort's age in the jump-off year as its *truncation age*: the last age at which its fertility is observed. For a cohort with truncation

age $a$, rates at ages 15 through $a$ are taken as observed and rates at ages $a+1$ through 40 are forecast, so its forecast CCF is the sum of an observed and a forecast piece. Truncation age therefore determines the forecast horizon: a cohort truncated at 39 contributes a single forecast age, while one truncated at 20 contributes twenty. Following the reference paper, evaluation is restricted to truncation ages 20-39.

## 2.3 Models

### 2.3.1 Benchmark models

This paper compares a simple Multi-Layer Perceptron (MLP) architecture to the best cohort fertility forecasting models as identified by Bohk-Ewald et al. (2018). I select the top 6 models as ranked by their statistical criteria and compare performance of these models to an out-of-the-box MLP model, named the Completed Cohort Fertility Neural Network or CCF-Net, trained to minimize mean squared error on age-specific fertility rates.

All benchmarks are re-implementations or direct adaptations of the reference code and/or written instructions accompanying Bohk-Ewald, Li and Myrskylä (2018). All methods producing period rate forecasts are converted to cohort rates as a last step in the forecasting process. More detailed descriptions of each model can be found in the supplement to the reference paper.

**Freeze rates** This model serves as the naive baseline. Each age-specific period rate is held constant at its last observed value from the jump-off year onward.

**Lee (1993)** A Lee–Carter model fit to log ASFR using the demography package (`demography::lca`), with the mortality-specific adjustment disabled (`adjust = "none"`), the jump-off set to the actual observed rates (`jumpchoice = "actual"`), and the time index k(t) extrapolated by an ARIMA model. Rates of 0 are set to $10^{-5}$ before taking logs and forecast rates are capped at 1.0, since the extrapolation of k(t) can otherwise diverge to demographically impossible values. Following the best-performing setting in the reference study, the model is fit on a fixed 40-year observation window ending at each jump-off year.

**Myrskylä et al. (2013)** A model that freezes the rate of change in fertility. For each age, the level $a$ is the rate in the jump-off year and the slope $b$ is the average annual change over the last five observed years; forecast rates are $a + b\delta_t$, where $\delta_t$ increments annually for the first five forecast years and is then held constant, so the recent trend is extrapolated for a limited period and frozen thereafter. Negative forecast rates are truncated at zero. The model is fit over all countries jointly so that the prediction-interval variance is pooled across the sample, as in the original implementation, but the point forecasts do not depend on that pooling.

**de Beer (1985, 1989)** An age-cohort time-series model where changes in fertility between ages and cohorts are modeled with two integrated ARIMA models. I use the best variant from the reference paper with ARIMA orders (1, 0, 0) across age and (2, 1, 0) across cohort, estimated by nonlinear least squares (`minpack.lm`). The model is fit on a 40-year observation window ending at each JOY. For five jump-off years in Japan (1986-1990), de Beer errors explode due to explosive auto-regressive coefficients. These years are dropped from the common evaluation pool.

**Schmertmann (2014)** A Bayesian model over the age × cohort fertility surface, forecast as a posterior mean under shape and time-series penalties. Each forecast surface covers 40 birth cohorts (JOY - 54 ... JOY - 15) across ages 15–40. Shape and time-series priors are calibrated on the completed fertility of all HFD countries. This is the only model fed vertical parallelogram cohort data (ASFRVV) rather than a period-to-cohort approximation, as is the case in the reference paper implementation (Bohk-Ewald et al., 2018). This means that performance scoring for Schmertmann (2014) is based on a different actual rate set than any of the other methods. I retain the reference implementation here, while acknowledging the comparability concerns it raises.

**Ševčíková et al. (2016)** The UN Bayesian approach. Applies a Bayesian hierarchical model to forecast TFR in a first step. In a second step, TFR forecasts are split into age-specific components by deriving proportionate ASFRs using a blend of an extrapolation of the country of interest's trend and a global pattern. The final proportionate ASFRs are a weighted average of the two trends. A few adaptations are made to the original Ševčíková et al. (2016) paper, in line with Bohk-Ewald et al. (2018). These include modeling by single year of age rather than 5 year intervals and using the HFD pool of countries rather than 200 UN countries.

The benchmark models fall into two primary categories: extrapolation-based and Bayesian. While all methods rely on extrapolation to some degree, purely extrapolation-based approaches (de Beer, 1985, 1989; Lee, 1993; Myrskylä et al., 2013) assume that a relatively simple, country-specific pattern of age-specific fertility persists over time. In contrast, Bayesian approaches (Schmertmann et al., 2014; Ševčíková et al., 2016) leverage partial pooling and informative priors to share strength across contexts and embed structural domain knowledge into the model. Bayesian approaches also inherently produce posterior distributions rather than single point predictions.

Neural networks integrate elements of both paradigms—extrapolating trends while pooling cross-population information—but substitute data-driven representations for theory-driven priors. Their non-linear architecture allows them to discover complex patterns that rigid theoretical priors might overlook or fail to parameterize.

Bohk-Ewald et al. (2018) find that the Bayesian approaches (Schmertmann et al., 2014; Ševčíková et al., 2016), despite incorporating cross-population pooling and structural priors, do not outperform the top-performing extrapolation methods, demonstrating that added model complexity does not inherently translate to superior predictive accuracy. This study investigates whether the distinct form of complexity introduced by neural networks—extensive representational capacity to learn nuances in the data—can overcome these limitations to improve forecast performance.

## 2.4 CCF-Net: proposed neural network

Figure 1 describes the architecture of CCF-Net: an ensembled Multi-Layer Perceptron (MLP) trained jointly on all admissible countries, so that information is pooled across populations rather than fitting each country in isolation[1]. The structure of this model draws on Richman and Wüthrich (2021)'s MLP for mortality forecasting, but is scaled down for this

[1]Key deep learning terms used in this section can be found in Appendix section A

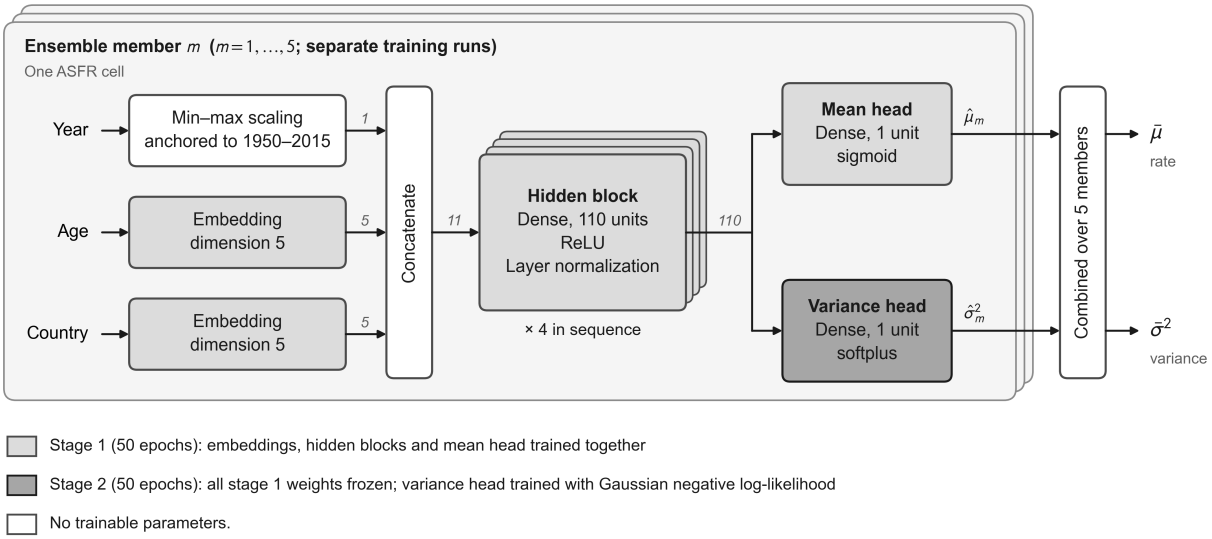



Figure 1: Diagram describing the architecture of CCF-Net. Each ASFR cell is described by year, age, and country. The concatenated input vector passes through four hidden blocks of 110 nodes. A sigmoid activation function on the first head's final layer gives the predicted fertility rate. After the weights are frozen, a softplus head is trained to predict the variance. Predictions are combined across five separately trained members. Grey italicized numbers give the vector dimension at each step.

data. Model inputs are calendar year, age, and country. Year enters as a continuous feature under a min-max transform; forecast years extrapolate linearly beyond this range. Age and country enter as learned embeddings (embedding layers of dimension 5). The model outputs a fertility rate and an associated variance. Because year enters continuously, the network extrapolates in calendar year when forecasting, continuing age-country specific trends. Each model is trained on 40 years of data, in keeping with several of the benchmark models.

The design of CCF-Net also builds on deep ensembles, one of the most successful approaches to uncertainty quantification in deep learning (Lakshminarayanan et al., 2017). In the original formulation, 5–10 networks are trained independently, each predicting both a mean and a variance, with every layer optimized end to end under a Gaussian negative log-likelihood (NLL). CCF-Net instead separates the two tasks. Nearly all of the network's capacity is devoted to the mean, which is trained on mean-squared error; the variance is then learned by a single additional layer, trained under the Gaussian NLL while the rest of the network is held fixed. This split follows an approach that has worked well in mortality forecasting (Park, 2026). The cost is that the variance must be inferred from representations learned for the mean, which likely sacrifices some accuracy in the variance estimates. Even so, the resulting uncertainty intervals remain reasonable (Figure A1).

For each ensemble member, there are two 'heads,' or components, to the model. The first head establishes the point prediction. In it, the concatenated feature vector passes through four hidden blocks of 110 ReLU units, each followed by layer normalization and mapped to a single output through a sigmoid activation, which constrains predicted rates to (0, 1).

This head is trained to optimize mean-squared error. The weights of the network are then frozen, and a second head—a single linear unit attached to the final hidden layer and passed through a softplus function—is trained to predict the variance of each rate using a Gaussian negative log-likelihood loss. This likelihood uses the frozen predicted rate as its mean.

The mean and variance values produced by each ensemble member are combined in the following ways to produce the final forecasts and variances. The forecasted rate is the simple average of the means produced by each ensemble member.

$$\bar{\mu} = \frac{1}{M} \sum_{m=1}^{M} \hat{\mu}_m(x) \tag{1}$$

The ensembled variance estimate is calculated by summing the mean of each member's predicted variance and the calculated variance of the five mean predictions:

$$\bar{\sigma}^2 = \frac{1}{M} \sum_{m=1}^{M} \hat{\sigma}^2 + \frac{1}{M} \sum_{m=1}^{M} (\hat{\mu}_m(x) - \bar{\mu}(x))^2 \tag{2}$$

Variation between members is driven by random initialization and batch draws on the full training data.

I do not tune CCF-Net's hyperparameters. The aim of this paper is not to identify the best possible neural architecture for fertility forecasting, but to test whether a simple, conventionally configured network can compete with established methods. All architectural and training choices (number of layers and nodes, embedding dimension, learning rate, batch size, and number of epochs) were fixed before any forecasts were evaluated, using common defaults for data of this size and a scaled-down version of Richman and Wüthrich (2021)'s architecture. Because nothing is tuned, no separate validation set is required.

Forgoing a conventional validation–test split is also a deliberate choice for this problem. Forecast accuracy is highly sensitive to the jump-off year: a method that performs well from one jump-off year may perform poorly from another. Tuning on one period and testing on a single held-out period would make both the selected configuration and the reported accuracy depend on which years happen to fall in each split. Instead, I follow Bohk-Ewald et al. (2018) and evaluate performance across the full set of HFD jump-off years, which keeps the results directly comparable to their benchmarks. Because Bohk-Ewald et al. (2018) select among model variants using these same jump-off years, the benchmark methods benefit from a degree of selection that CCF-Net does not.

As a check on the sensitivity of these hyperparameters, I train 5 additional variants after the main analysis, varying dropout, number of years of training data, and model dimensions (Appendix Table A2). Most variants are competitive with the top-performing benchmarks and all outperform freeze rates. One variant is ranked higher than CCF-Net on the Bohk-Ewald et al. (2018) metrics, a larger model, but CCF-Net is the second best of the variants tested.

## 2.5 Evaluation

Models are evaluated according to the criteria established in Bohk-Ewald et al. (2018). Methods are ranked by the Kolmogorov-Smirnov (KS) test result, where ties are broken

by mean absolute percentage error (MAPE). Absolute percentage error is calculated on completed cohort fertility (the sum of age-specific fertility rates (ASFRs) of one birth cohort from 15 to 40, abbreviated CCF) and is calculated with the following formula:

$$\frac{|CCF_forecast - CCF_observed|}{CCF_observed} * 100$$

For every pair of methods A and B, I perform the one-sided two-sample KS test on their distributions of absolute percentage errors (APEs) in completed cohort fertility. The test statistic is the largest gap by which A's empirical cumulative density function (CDF) exceeds B's. A significant result ($p < 0.05$) means that A's CDF lies to the left of B's somewhere, meaning that a larger share of its forecasts fall under a certain APE. Following Bohk-Ewald et al. (2018), the test is run from both sides (K(A, B) and K(B, A)). A dominates B if K(A, B) is significant and K(B, A) is not. If both tests are significant, neither dominates. The table reports how many other methods that method dominates.

Standard KS p-values assume independent observations. This assumption is violated here as errors are correlated within countries, across adjacent cohorts, and across jump-off years, resulting in p-values that are too small. Therefore, I obtain p-values from a paired permutation test. In each of 10,000 permutations, the two methods errors are swapped for all forecasts of a randomly chosen subset of countries and test statistics are recalculated. The p-value is then the share of permutations whose statistic is at least as large as the observed one.

Mean APE breaks KS test ties. MAPE averages across the errors produced by each method and smaller values imply better forecasts. Table 1 also reports APE thresholds, below which 50 / 80 / 90 / 95% of completed cohort forecasts fall. The last column reports root mean squared error (RMSE) on CCF.

Figures 2 and 5 also report RMSE, but on the age-specific rates rather than CCF. This allows us to look at error by forecast horizon. Figure 3 looks at the magnitude of the difference between forecasted fertility rates between CCF-Net and the benchmarks, establishing that on the evaluation periods tested CCF-Net predicts higher cohort fertility. Figure 4 compares CCF-Net and Myrskylä et al. forecasts of cohort fertility in each country. Finally, Figure 6 plots forecasts past the evaluation window, from the 2019 jump off year onward.

## 2.6 Uncertainty Quantification

In this paper, I am interested first and foremost in comparing the accuracy of each model's point predictions. However, I acknowledge the central importance of uncertainty intervals for making any credible predictions about the future. Appendix Section C contains a preliminary comparison of the prediction intervals generated by each approach, showing the empirical coverage of the 95% prediction intervals. While neural networks traditionally only produced point predictions, many methods have now been developed to incorporate uncertainty quantification (Abdar et al., 2021; He et al., 2026). This paper uses an approach designed to capture both aleatoric and epistemic uncertainty by first, having the network predict both a mean and variance for each fertility rate, rather than a point prediction, and second, by ensembling five separately trained networks. To get the 95% intervals, $1.96 * \bar{\sigma}$ is subtracted and added from each rate to get the lower and upper bounds.

# 3 Results

Table 1: Rank methods based on different measures of accuracy in forecasting completed cohort fertility (CF, ages 15–40). Errors are absolute percentage errors (APEs) of forecast CF, computed on an equal testing dataset of 11,599 CF forecasts (32 countries, truncation ages 20–39, jump-off years 1985–2013). KS is the number of other methods a method strictly stochastically dominates (one-sided two-sample Kolmogorov–Smirnov statistic, $P = 0.05$, crossovers excluded), with P values from a paired permutation test that swaps the two methods' errors country by country (10,000 permutations), so that errors of the same country are not treated as independent; the remaining columns are in % APE. The overall ranking in column 1 is based on the KS test statistic (column 2), ties broken by the overall mean (column 7). Within-column ranks are in parentheses; the last row counts how many method pairs each column orders differently from the overall ranking.

| | | Threshold APE, % | | | | | |
|---|---|---|---|---|---|---|---|
| Overall ranking | KS | 50 | 80 | 90 | 95 | Mean | RMSE |
| 1. CCF-Net | 5 (1) | 1.16 (2) | 5.18 (2) | 8.85 (3) | 12.41 (3) | 3.02 (1) | 5.24 (1) |
| 2. Myrskyla et al. | 5 (1) | 1.07 (1) | 5.16 (1) | 9.08 (4) | 13.31 (5) | 3.14 (3) | 5.76 (4) |
| 3. Schmertmann et al. | 4 (3) | 1.29 (3) | 5.18 (3) | 8.38 (1) | 11.66 (1) | 3.09 (2) | 5.41 (2) |
| 4. de Beer | 3 (4) | 1.35 (4) | 6.08 (5) | 9.80 (5) | 13.53 (6) | 3.44 (5) | 5.94 (5) |
| 5. Sevcikova et al. | 1 (5) | 1.66 (5) | 5.49 (4) | 8.45 (2) | 12.21 (2) | 3.31 (4) | 5.52 (3) |
| 6. Freeze rates | 0 (6) | 2.16 (7) | 7.23 (6) | 10.24 (6) | 13.25 (4) | 3.99 (6) | 6.24 (6) |
| 7. Lee | 0 (6) | 1.81 (6) | 7.39 (7) | 11.63 (7) | 16.13 (7) | 4.19 (7) | 7.07 (7) |
| No. of inversions | 0 | 2 | 2 | 5 | 7 | 2 | 3 |

Following the ranking system established in Bohk-Ewald et al. (2018), I find that CCF-Net is ranked first, though narrowly, when performance is averaged across JOYs 1985-2013 (Table 1). Its performance is tied with Myrskylä et al. (2013) on the KS test for stochastic dominance, but the tie is broken by the mean absolute percentage error. CCF-Net has the lowest RMSE on completed cohort forecasts. While the margin of improvement is fairly small, this result demonstrates that a simple off-the-shelf ensembled neural network performs at least as well on this problem as the very best approaches in demography.

Figure 2 shows that CCF-Net performs best for more countries than any other method (13/31), and does particularly well relative to the baselines for many post-socialist countries (Russia, Belarus, Ukraine, Slovakia, Estonia, Bulgaria). CCF-Net also gets progressively better than other methods at longer forecast horizons (years past jump-off).The curved pattern in Figure 2b is due to the correlation between longer forecast horizons and older age predictions. Fertility is most predictable (low RMSE) at the youngest and oldest ages, where fertility is low and least predictable in the middle of the childbearing window.

Compared with the benchmarks, CCF-Net tends to predict higher completed cohort fertility in these evaluation periods, which turns out to be more accurate on average (Figure 3). All approaches under-forecast, but CCF-Net does so to a lesser extent. On average, CCF-Net predicts a higher number of children per woman on the historical time periods

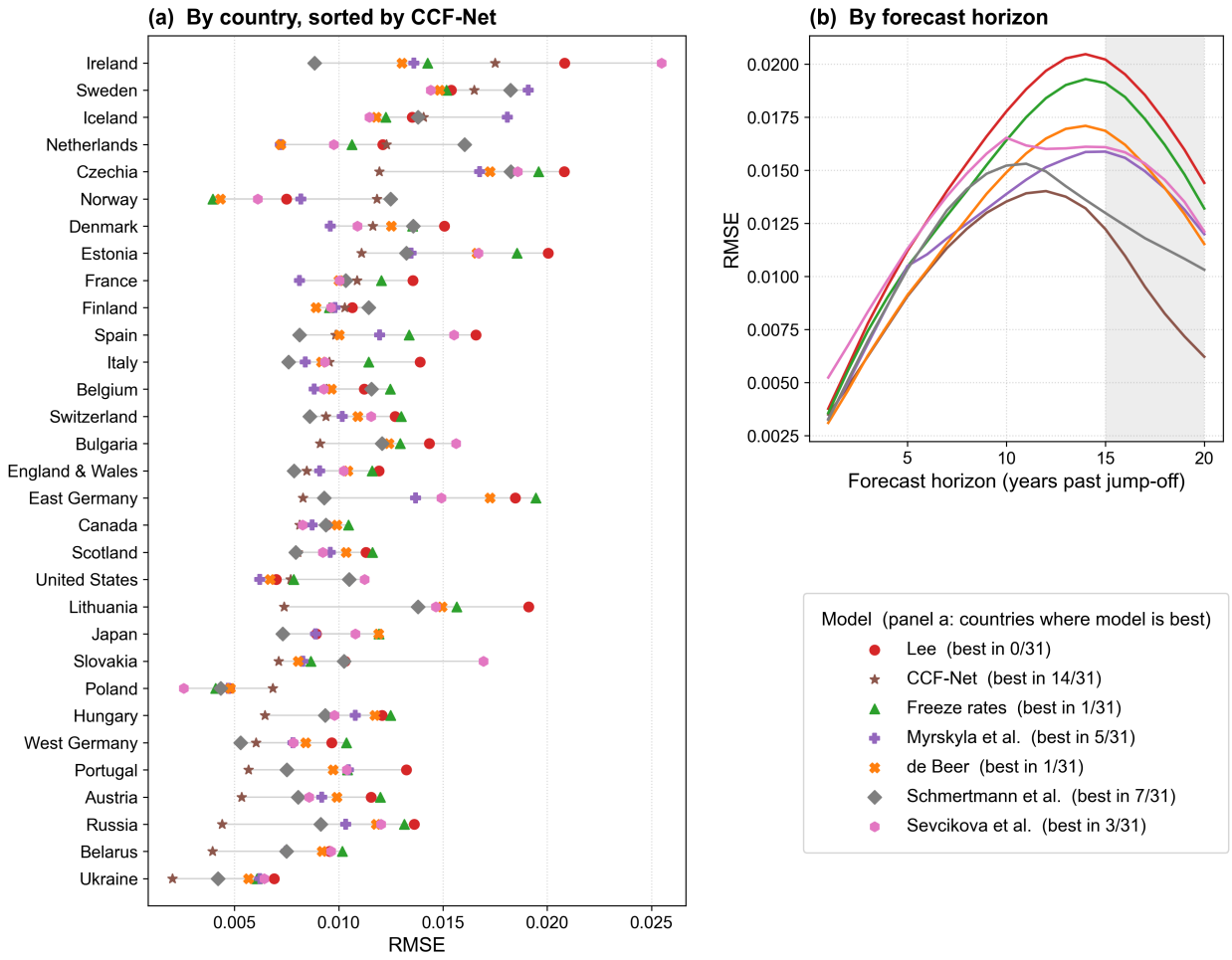



Figure 2: RMSE on age-specific rates for completed cohorts by country and by forecast horizon. Legend reports the number of countries for which a model is the best performing model. Only countries with more than three jump off years are included in the figure: 31 of the total 32 countries evaluated. The shaded region of (b) represent RMSEs for fewer forecasts as not all JOYs have 15-20 year horizons completely observed yet.

tested than the other methods, which is also closer to observed CCF. The most notable exception is Ševčíková et al. forecasts, which are higher than CCF-Net forecasts (and closer to observed CCF) when forecasting from the youngest ages (20-24). This reverses from age 24-39.

Figure 4 plots observed completed cohort fertility alongside CCF-Net and Myrskylä et al. forecasts at the 2005 jump off year. 2005 is selected to showcase one of the jump off years where the deep learning model does quite well relative to the other models. The Myrskylä model is selected as the best performing comparator. In many countries, particularly Western European countries (e.g. England and Wales, the Netherlands, Norway, Scotland, Spain), the Myrskylä and CCF-Net forecasts look nearly indistinguishable. Countries with the most dramatic divergence include post-socialist Eastern European countries (e.g. Poland, Russia, Belarus, Lithuania) and Taiwan. In all of these cases, CCF-Net's higher prediction turned

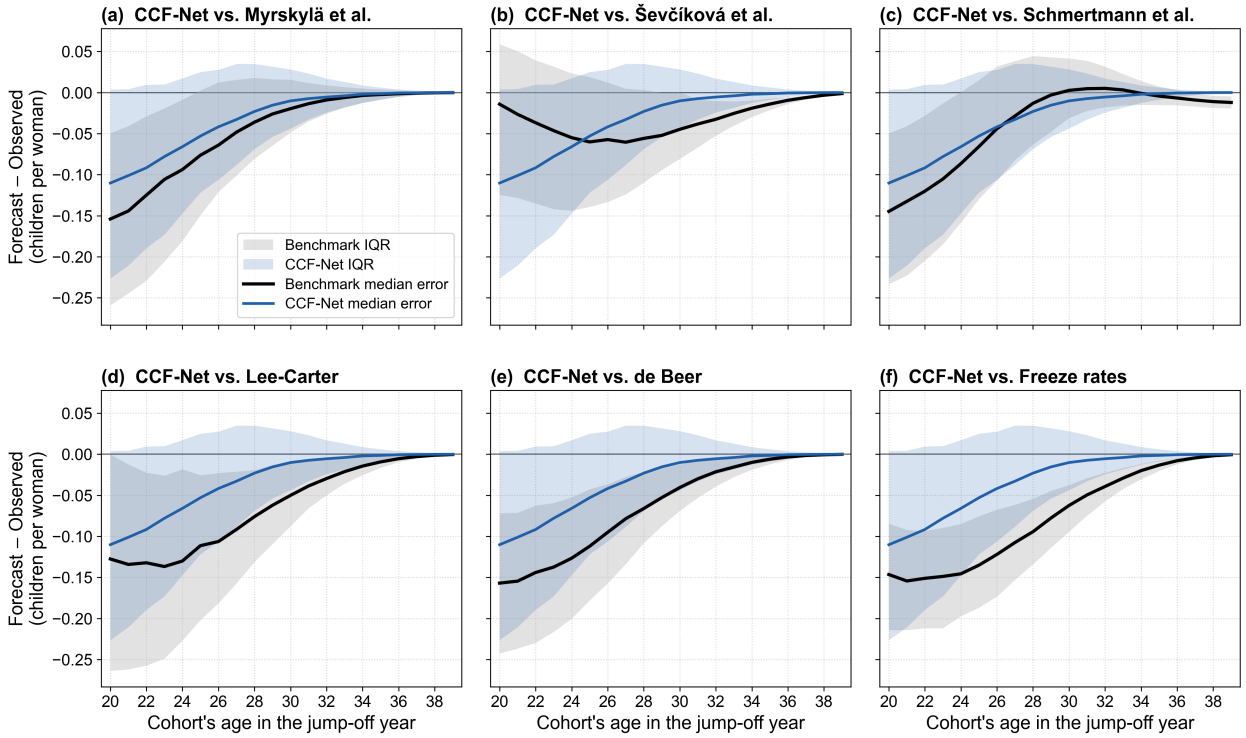



Figure 3: Difference between each method's median forecast of completed cohort fertility (children per woman) by the cohort's age in the jump-off year and the observed median for that age. CCF-Net forecasts are visualized in every panel to compare with the benchmark. The shaded region represents the inter-quartile range of differences for each method, while lines represent the median forecast differences.

out to be more accurate. There are also a couple cases where CCF-Net overshoots and the lower Myrskylä forecast is closer to the truth (e.g. United States, Finland).

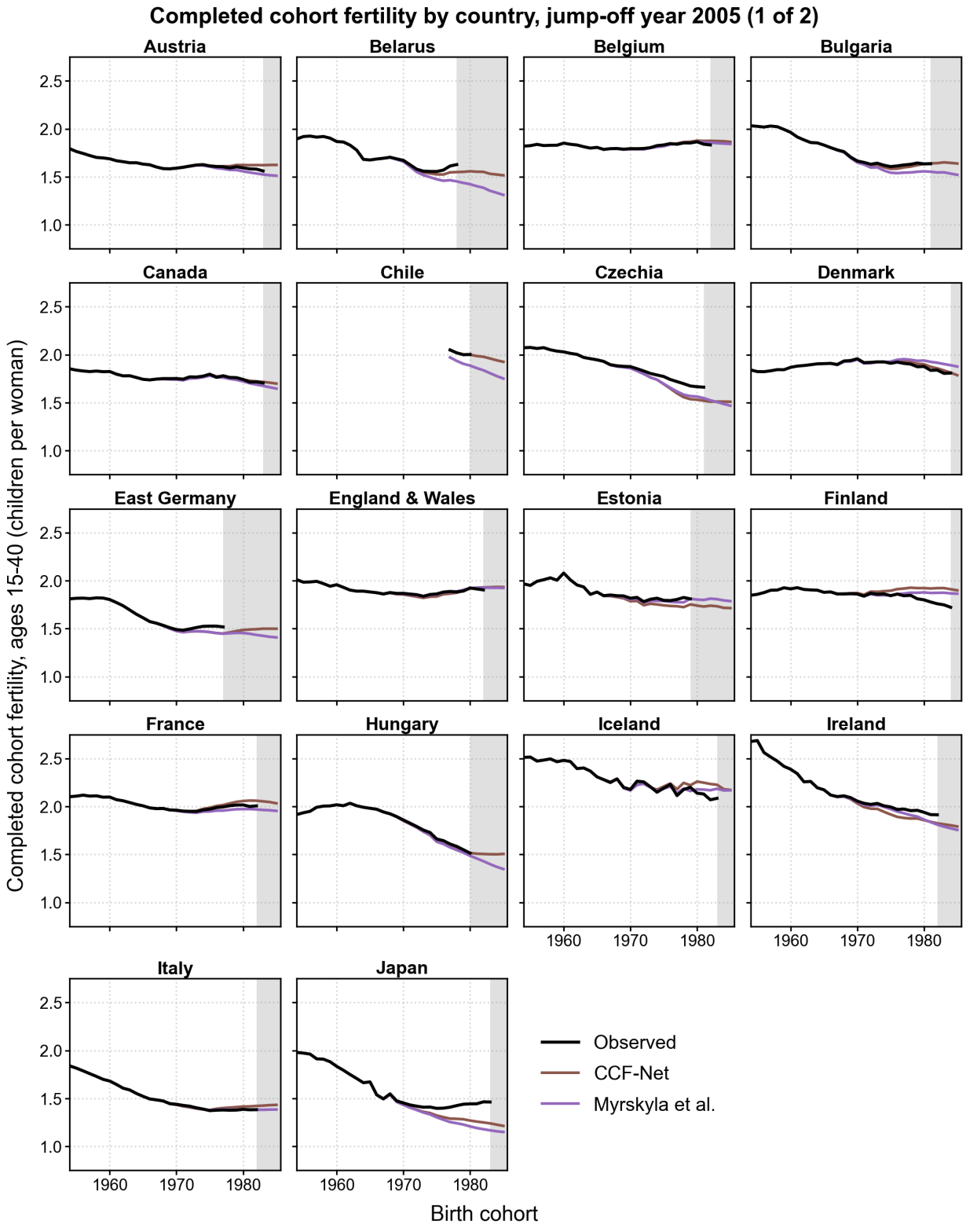
While CCF-Net performs best on average, one caveat is that it does very poorly relative to other models in several of the more recent jump-off years: 2006-2011 (Figure 5). It seems that CCF-Net's tendency to predict higher fertility than other models harms predictions, at least in 2010 (Figure A2). In its training data, cohorts that postponed childbearing largely made up for it at older ages (Goldstein et al., 2009); in some countries, this recuperation weakened from the late 2000s onward, so that births were "not just later, but fewer" (Hellstrand et al., 2021). It seems that CCF-Net was slower to adjust to this reduction in fertility. A second caveat, also visible in Figure 5, is CCF-Net's volatility. Whereas the RMSE of the other approaches varies smoothly across jump-off years, CCF-Net's fluctuates considerably: a prime demonstration is CCF-Net going from the best performer in 2005 to the worst in 2008, then back to the best in 2012.

When trained on data through 2019, the last year with available data for most countries, CCF-Net predicts substantially *lower* completed cohort fertility, on average, than the other approaches. This suggests it has picked up the recent shift toward falling or stagnating cohort fertility that began around 2010 in many countries. Figure 6 shows a precipitous drop in CCF-Net's predicted completed fertility, one that diverges sharply from the other methods. Combined with CCF-Net's worse performance in more recent years (2006-2011), this extremely low CCF of 1.4 seems unrealistic. Part of the issue here and with CCF-Net's variable performance may be the lack of structure in the MLP design. Architectures more capable of explicitly considering the time series nature of the data, like RNNs and LSTMs, may be able to produce more stable long run predictions.

# 4 Conclusion

Deep learning models produce accurate forecasts of age-specific mortality (Zheng et al., 2025) and of the TFR (Ciganda et al., 2026). This paper asked whether the same holds for completed cohort fertility. Using the models, data, and evaluation criteria from Bohk-Ewald et al. (2018), the study that established demography's informal "leaderboard" for CCF accuracy, I find that a simple Multi-Layer Perceptron, used largely off-the-shelf, achieves lower average forecast error than the methods evaluated in that study. These findings indicate that neural networks can forecast CCF with accuracy competitive with established methods, though it remains to be seen whether this advantage holds in future evaluation periods.

My results are consistent with findings on neural networks in mortality forecasting (Zheng et al., 2026). In particular, this paper uses a fully connected feed-forward network (FFN) or multi-layer perceptron (MLP) modeled after the architecture of Richman and Wüthrich (2021). As in their study, this FFN outperforms classic benchmarks on predictive accuracy of age-specific rates. Here I show that this also translates to improved accuracy for aggregated completed cohort fertility. CCF-Net's advantage over classic approaches is largest in many post-socialist Eastern European countries, a pattern also observed for deep learning models of mortality (Park, 2026). The results also parallel work on TFR forecasting, which finds that a neural network outperforms other sophisticated approaches, including Bayesian models, on point forecasts (Ciganda et al., 2026). Taken together, these studies suggest that the


Completed cohort fertility by country, jump-off year 2005 (1 of 2)
Austria
Belarus
Belgium
Bulgaria
Canada
Chile
Czechia
Denmark
East Germany
England & Wales
Estonia
Finland
France
Hungary
Iceland
Ireland
Italy
Japan
Completed cohort fertility, ages 15-40 (children per woman)
2.5
2.0
1.5
1.0
1960
1970
1980
Birth cohort
Observed
CCF-Net
Myrskyla et al.

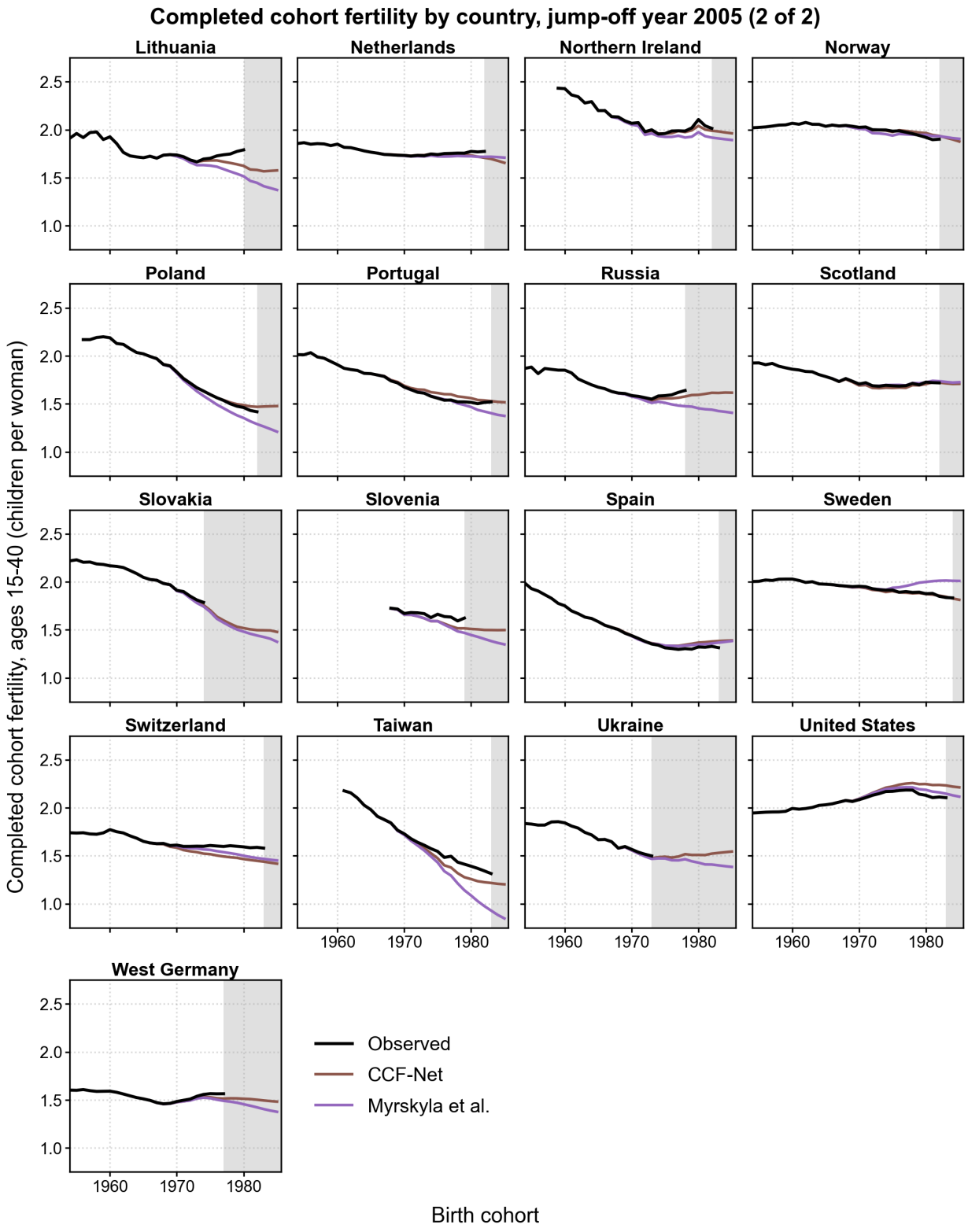



Figure 4: Completed cohort fertility by country from jump off year 2005. The black line depicts observed completed cohort fertility while the brown and purple lines represent CCF-Net and Myrskylä forecasts. The gray panel represents cohorts whose fertility data is not completely available yet in the HFD and thus cannot produce an observed value for completed cohort fertility.

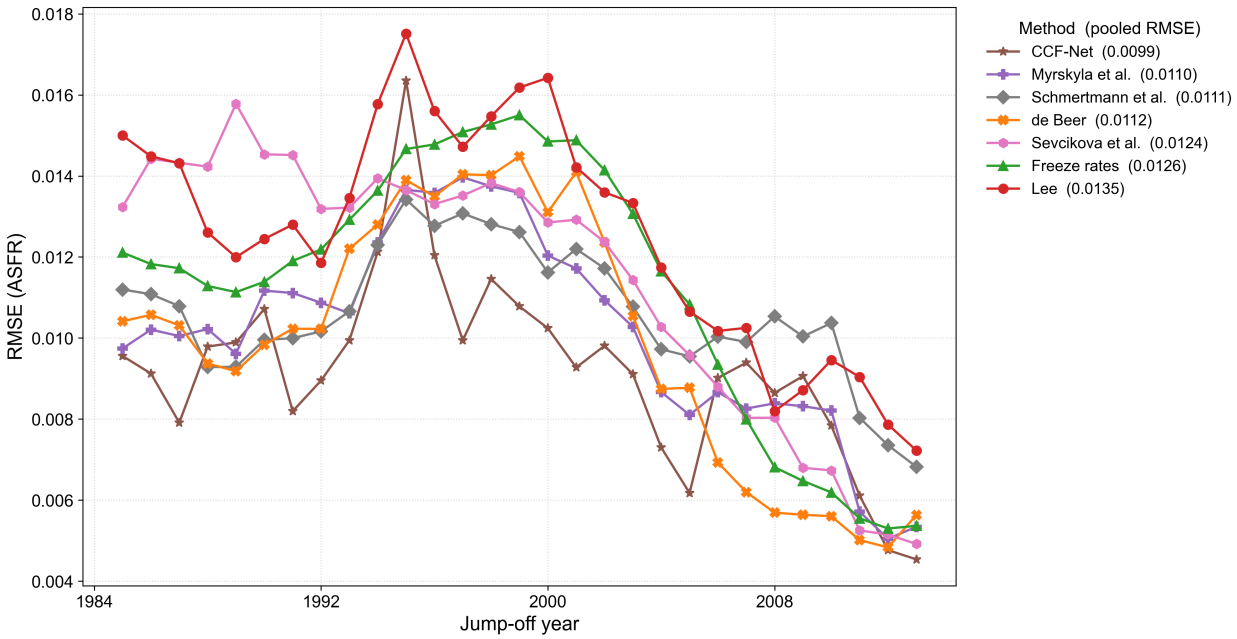



Figure 5: Pointwise RMSE by jump-off year on completed-cohort forecast ASFR cells from the complete set of CCF forecasts (JOYs 1985-2013, truncation ages 20-39).

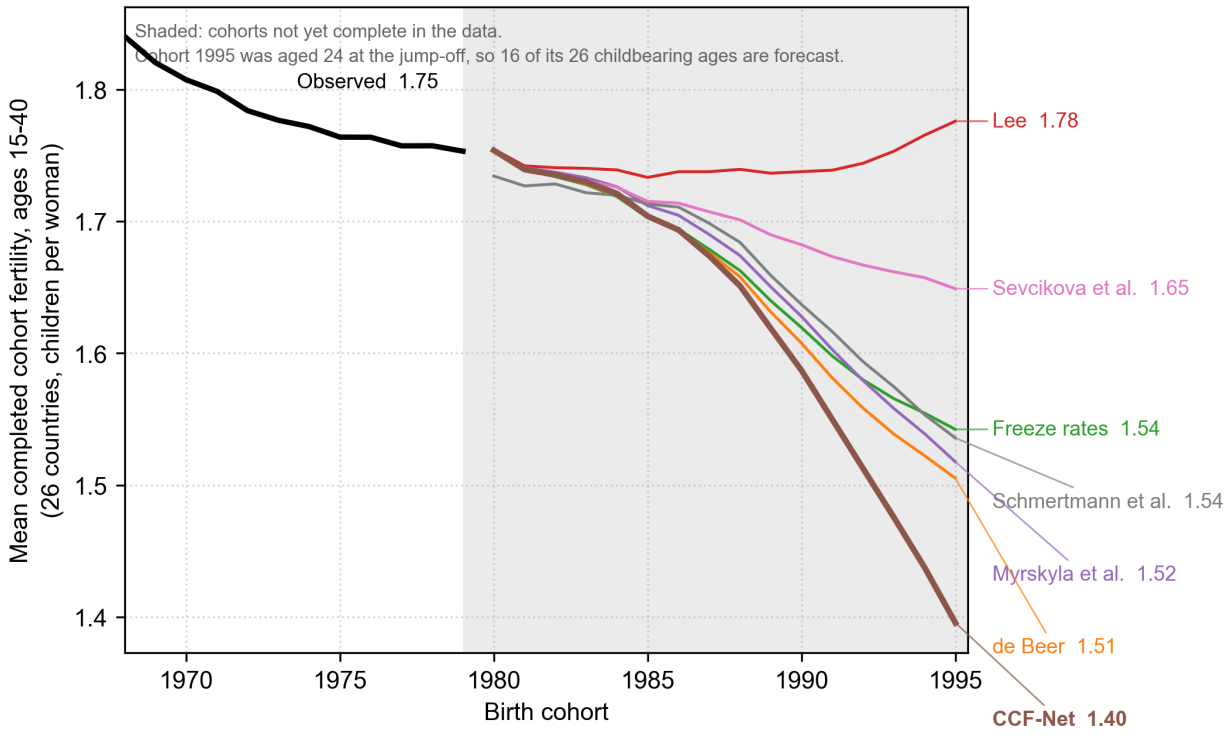



Figure 6: Mean of completed fertility predictions across the 26 countries with 40 years of complete data through 2019.

accuracy gains from deep learning are not confined to a single demographic outcome; this paper adds cohort fertility to that list.

The results are somewhat surprising given a commonly cited limitations of deep learning: these models often underperform when data are scarce. Fertility data are limited, with short time series and a narrow range of ages. With pooling, however, the model has tens of thousands of observations (age-, country-, and year-specific fertility rates) to train on per jump-off year. This is small by machine learning standards (Grinsztajn et al., 2022), yet CCF-Net outperforms every benchmark tested.

Substantively, CCF-Net's forecasts are not merely more accurate but systematically higher in the evaluation period, particularly in post-socialist countries, where CCF-Net better anticipates fertility rises observed in many countries. These higher predictions hurt its performance in JOY 2010, where observed cohort fertility in many countries begins to decline or stagnate (Figure A2). When trained on data up through 2019, CCF-Net carries the decline forward, and more dramatically than other methods. Ciganda et al. (2026) also find that their deep learning approach predicts greater declines in TFR than the Bayesian comparator, but that declines are less extreme than those anticipated by Global Burden of Disease (GBD) project forecasts. Here, the deep learning approach produces the lowest CCF forecast of all methods compared, and by a large margin. Whether declines of this magnitude materialize cannot yet be evaluated.

CCF-Net's instability over time (Figure 5) and its tendency toward extreme forecasts (it often produced the highest or lowest forecast of the models tested) are serious limitations. A model that is the most accurate on average but sometimes produces forecasts far from the truth is hard to rely on in practice, especially when the way it arrives at those forecasts is opaque. This instability may come from CCF-Net's specific architecture (an MLP), or it may be a property of neural networks in general. The flexibility that allows the neural networks to learn non-linear relationships may extrapolate less predictably than more constrained models. Testing other architectures could help determine whether this is a fundamental issue with neural networks or whether it can be resolved by a model with additional structure.

Alongside tests of other architectures, future research could carry out more systematic hyperparameter tuning to arrive at the optimal model for this problem. I do not carry out robust hyperparameter tuning to arrive at the best deep learning architecture. This is partly to match the style and comparison set of the reference paper, and partly because the paper asks a deliberately modest question: can a deep learning model succeed here with no tuning at all? Five alternative specifications are reported in Appendix Table A2. If machine learning models increasingly become a part of the demography toolkit, it would be wise to develop more robust model testing procedures that include tuning on a separate validation set and retaining a held-out set for final analysis.

Uncertainty intervals are not the focus of this paper, and accordingly, comparisons of the coverage and width of intervals are only briefly considered in the appendix. Future work should carry out more robust comparisons of various methods prediction intervals. In addition, I use a single approach to calculating uncertainty, but future studies could compare multiple methods of neural network uncertainty quantification for this problem.

Further, this paper does not examine in depth why CCF-Net produces the forecasts it does. Given the model's opacity, interpretability methods could help identify which inputs drive its unusually high and low forecasts, and whether those drivers are demographically

plausible.

Finally, this analysis is restricted to data from the Human Fertility Database (HFD), which limits its geographic scope. These findings may not hold for other areas of the world with different demographic regimes, particularly countries in Africa, South Asia, and Latin America. Future work could extend the comparison using broader data sources such as the United Nations World Population Prospects (WPP), which provides estimates for nearly every country and territory in the world. Doing so would involve trade-offs as WPP estimates are partially model-based and sometimes not available in 1-year age increments. Nevertheless, applying this analysis to a more diverse set of countries would help to establish how well CCF-Net, or other neural networks, generalize.

This paper provides evidence that neural networks are a viable approach to forecasting completed cohort fertility, with accuracy competitive with established methods. Deep learning methods may not yet be practical for most demographers, and their adoption by forecasting institutions such as the UN may depend on wider computational training in the field and better-established approaches to uncertainty quantification. As these methods mature, they may warrant consideration alongside established methods for fertility and mortality projection.

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# Appendix for "Can a Simple Neural Network Improve Cohort Fertility Forecasts?"

# A Deep learning glossary

Table A1: Glossary of deep learning terms used in the description of CCF-Net. *First draft of glossary was created with Claude Opus 5.5 but was reviewed and edited by the author for accuracy and clarity.*

| **Term** | **Definition** |
|---|---|
| *Architecture* | |
| Neural network; Multi-Layer Perceptron (MLP) | A flexible nonlinear regression function built by composing simple transformations ("layers"). Each layer takes linear combinations of the previous layer's outputs and passes them through a fixed nonlinear function. In an MLP, every output of one layer feeds into every unit of the next, and information flows in one direction from inputs to output. |
| Weights (and biases) | The parameters of the network: the coefficients and intercepts of the linear combinations in each layer. "Training" a network means estimating these parameters by numerically minimizing a loss function. |
| Feature | An input variable to the model; equivalent to a covariate or predictor. |
| Layer; hidden layer | One stage of transformation in the network. The input layer receives the features, the output layer produces the prediction, and the intermediate ("hidden") layers produce derived quantities that are not directly observed. |
| Hidden block | A hidden layer together with the operations applied to its output (here, a ReLU activation followed by layer normalization). |
| Unit; node | A single element of a layer: one linear combination of the previous layer's outputs, followed by an activation function. A layer of 110 units produces 110 derived variables for each observation. |
| Activation function | A nonlinear function applied to each unit's linear combination. Without it, any stack of layers would collapse into a single linear model. Its role is loosely analogous to the (inverse) link function in a generalized linear model. |
| ReLU (rectified linear unit) | The activation function $f(z) = \max(0, z)$: it passes positive values unchanged and sets negative values to zero. |
| Sigmoid | The activation function $f(z) = 1/(1 + e^{-z})$, i.e., the inverse logit. It maps any real number into $(0, 1)$, which is why it is used to keep predicted fertility rates in that range. |

*Continued on next page*

*Table A1 continued*

| Term | Definition |
|---|---|
| Softplus | The activation function $f(z) = \log(1 + e^z)$, a smooth function that is always positive. It is used here to guarantee that predicted variances are positive. |
| Layer normalization | An operation that, for each observation, standardizes a layer's outputs across its units (subtracting their mean and dividing by their standard deviation) and then applies a learned rescaling and shift. It keeps the scale of intermediate values stable, which makes training more reliable. |
| Embedding | A learned numerical representation of a categorical variable. Each category (e.g., each country or each age) is assigned a vector of real numbers (here, of length 5) that is estimated jointly with the rest of the network. It is an alternative to dummy coding; categories that behave similarly tend to end up with similar vectors. |
| Representation | The values a set of hidden layers produces for a given observation; in effect, derived variables that the network constructs for itself during training. |
| Head | An output component attached to the shared hidden layers. A single network can have several heads, each predicting a different quantity from the same representation; CCF-Net has one head for the mean and one for the variance. |
| Linear unit | A unit with no nonlinear activation: its output is simply a linear combination of its inputs plus an intercept. |
| *Training* | |
| Loss function | The objective function minimized to estimate the weights, such as mean-squared error or a negative log-likelihood. |
| Gaussian negative log-likelihood (NLL) loss | The negative log of a normal density evaluated at the observed value, with mean and variance supplied by the network. Minimizing it is equivalent to maximum likelihood estimation of a heteroskedastic normal model in which both the mean and the variance depend on the inputs. |
| End-to-end training | Estimating all of a network's weights jointly under a single loss function, as opposed to estimating parts of the network in separate stages. |
| Freezing weights | Holding some parameters fixed at their current estimates while others continue to be estimated; similar in spirit to two-stage estimation. |
| Optimizer | The iterative numerical algorithm used to minimize the loss. Neural networks are usually trained with variants of stochastic gradient descent, which repeatedly nudges the weights in the direction that reduces the loss on a small random subset of the data. |
| Adam | A widely used optimizer. It is a variant of stochastic gradient descent that adapts the step size for each parameter using running averages of past gradients and squared gradients. |

*Continued on next page*

*Table A1 continued*

| Term | Definition |
|---|---|
| Learning rate | The step size of the optimizer: how far the weights are moved at each update. Too large a value makes training unstable; too small makes it slow. |
| Batch; batch size | The random subset of training observations used to compute each update of the weights, and the number of observations in it (here, 256). |
| Step | A single update of the weights, computed from one batch. |
| Epoch | A unit of training length. Conventionally, one full pass through the training data. |
| Random initialization | Weights are set to random starting values before training. Because the loss surface has many local minima, different starting values (and different random batch draws) lead to different fitted networks. |
| Dropout | A regularization technique that randomly sets a fraction of units to zero at each training step to discourage overfitting. Not used in CCF-Net. |
| Early stopping | A regularization technique that halts training once performance on held-out validation data stops improving. Not used in CCF-Net. |
| Hyperparameter | A setting chosen by the analyst rather than estimated from the data, such as the number of layers, units per layer, embedding dimension, learning rate, batch size, or number of epochs. Analogous to a tuning parameter such as the penalty in ridge or lasso regression. |
| Hyperparameter tuning | Selecting hyperparameters by comparing the performance of different settings on a validation set that is separate from both the training and test data. |
| *Ensembling and uncertainty* | |
| Ensemble; ensemble member | A collection of models whose predictions are combined, typically by averaging. Each individual model is a member; in CCF-Net, each member is a separately trained network. |
| Deep ensemble | The approach of Lakshminarayanan et al. (2017): several networks are trained independently from different random initializations, each predicting a mean and a variance. Disagreement between members captures uncertainty about the model itself, while the predicted variances capture noise in the data. |

# B Hyperparameter sensitivity checks

While this paper does not employ formal hyperparameter tuning, I do run post-hoc variants of CCF-Net as a sensitivity check. The CCF-Net parameters were selected prior to seeing any other variant results based on what seemed reasonable for this problem and data, so we might expect it to do fairly well compared to less reasonable modeling choices. But it would be concerning if other slightly adjusted architectures or modeling choices produce

vastly different and worse results. Here, I report performance for a few other variants. These variants were trained on 6 jump-off years (1985, 1990, 1995, 2000, 2005, and 2010) rather than every jump-off year between 1985 and 2013.

This included training models with and without dropout, more or less provided data, and different model dimensions (nodes and layers). CCF-Net sets dropout to 0, uses 40 years of data, and uses 4 hidden layers with 110 nodes each. The variants described below alter each of these settings one at a time.

- Variant 1: Adds dropout of 0.1
- Variant 2: Provides the model unlimited data
- Variant 3: Provides the model 35 years of data
- Variant 4: Alters model dimensions (increase from 110 nodes to 256 nodes)
- Variant 5: Alters model dimensions (increase from 4 to 5 layers)

Of course these represent only a handful of the possible values for these hyperparameters and a subset of the possible knobs that could be tweaked in a neural network. Future work should carry out a more comprehensive tuning process, including testing different architectures (recurrent neural networks, transformers, etc.) to determine the best performing neural net.

Table A2: Replica of Table 1, with deep learning variants included. Tested on a sample of the completed cohorts from Table 1 (only 6 jump-off years). CCF-Net ranks second to Variant 4 (the model with more nodes per layer), but is one of the best performing deep learning models tested. Other variants are worse than Myrskylä et al., but all are better than freeze rates.

| | | Threshold APE, % | | | | | |
|---|---|---|---|---|---|---|---|
| Overall ranking | KS | 50 | 80 | 90 | 95 | Mean | RMSE |
| 1. Variant 4 | 10 (1) | 1.11 (1) | 5.03 (1) | 9.10 (6) | 13.39 (6) | 3.09 (2) | 5.60 (5) |
| 2. CCF-Net | 7 (2) | 1.22 (4) | 5.22 (3) | 8.22 (1) | 11.08 (1) | 2.95 (1) | 5.03 (1) |
| 3. Myrskylä et al. | 7 (2) | 1.14 (2) | 5.48 (6) | 9.36 (7) | 14.30 (10) | 3.28 (6) | 5.92 (7) |
| 4. Variant 2 | 7 (2) | 1.15 (3) | 5.65 (7) | 10.05 (9) | 14.47 (11) | 3.36 (8) | 5.97 (8) |
| 5. Schmertmann et al. | 5 (5) | 1.34 (6) | 5.19 (2) | 8.66 (3) | 12.41 (3) | 3.17 (3) | 5.57 (4) |
| 6. Variant 3 | 5 (5) | 1.32 (5) | 5.45 (5) | 9.08 (5) | 13.68 (8) | 3.32 (7) | 5.90 (6) |
| 7. Variant 5 | 4 (7) | 1.35 (7) | 5.84 (8) | 9.07 (4) | 12.75 (4) | 3.26 (4) | 5.52 (3) |
| 8. de Beer | 3 (8) | 1.44 (8) | 6.22 (9) | 9.95 (8) | 13.63 (7) | 3.52 (9) | 6.00 (9) |
| 9. Ševčíková et al. | 2 (9) | 1.68 (9) | 5.42 (4) | 8.58 (2) | 12.16 (2) | 3.28 (5) | 5.40 (2) |
| 10. Variant 1 | 1 (10) | 1.91 (10) | 6.67 (10) | 10.37 (10) | 13.72 (9) | 3.84 (10) | 6.19 (10) |
| 11. Freeze rates | 0 (11) | 2.26 (12) | 7.29 (11) | 10.46 (11) | 13.37 (5) | 4.07 (11) | 6.35 (11) |
| 12. Lee | 0 (11) | 1.95 (11) | 8.28 (12) | 13.10 (12) | 17.49 (12) | 4.64 (12) | 7.83 (12) |
| No. of inversions | 0 | 4 | 10 | 19 | 28 | 11 | 18 |

Table A3: CCF-Net ensemble members scored individually. Pointwise RMSE is on the 197,361 common-grid cells; CF columns are absolute percentage errors of completed cohort fertility on the equal testing dataset of 11,599 forecasts. The ensemble is the per-cell mean of the members above it; two benchmarks are shown for context. The individual members are not better than the baselines. It is the ensemble that wins.

| Series | RMSE | CF APE, % | | |
|---|---|---|---|---|
| | | median | mean | P95 |
| seed 0 | 0.01391 | 1.47 | 3.39 | 13.23 |
| seed 1 | 0.01472 | 1.39 | 3.65 | 14.91 |
| seed 2 | 0.01296 | 1.32 | 3.27 | 13.04 |
| seed 3 | 0.01530 | 1.47 | 3.82 | 15.66 |
| seed 4 | 0.01461 | 1.25 | 3.52 | 14.91 |
| **ENSEMBLE (mean of 5)** | 0.01242 | 1.15 | 3.02 | 12.19 |
| Schmertmann et al. | 0.01401 | 1.29 | 3.09 | 11.66 |
| Myrskylä et al. | 0.01421 | 1.07 | 3.14 | 13.31 |

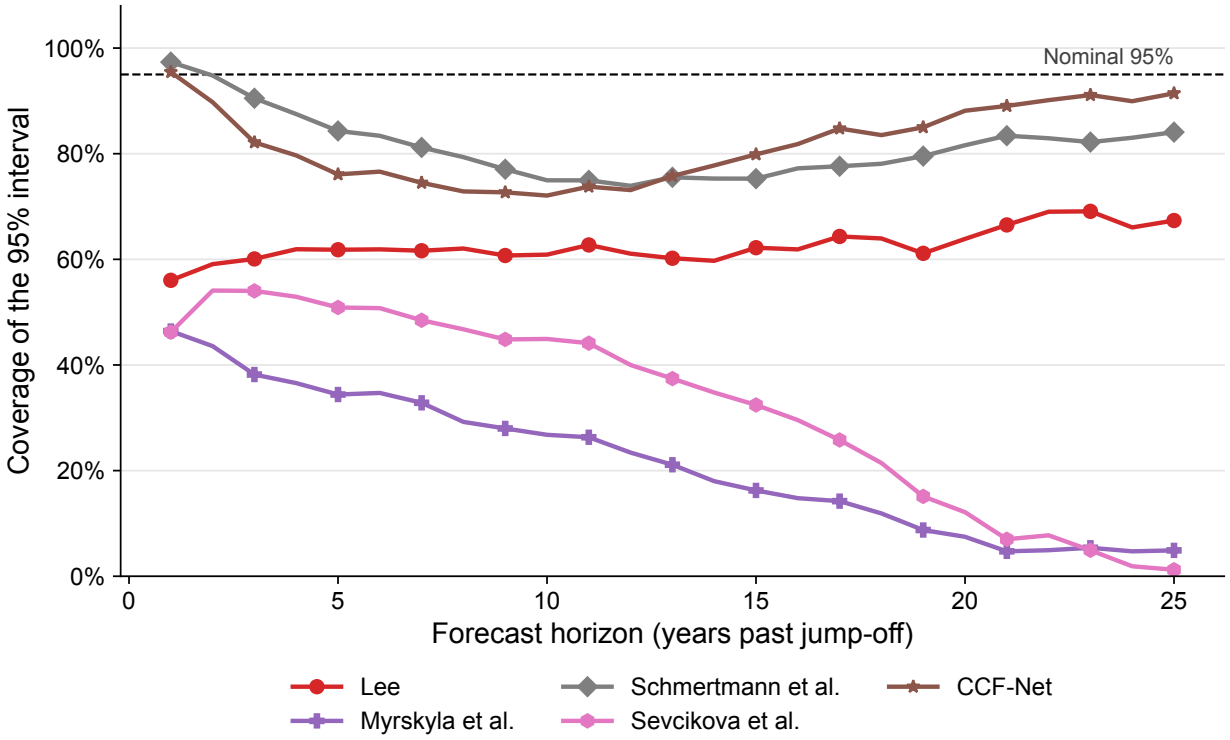



Figure A1: Empirical coverage of the 95% prediction intervals by forecast horizon.

# C Uncertainty Interval Evaluation

CCF-Net's uncertainty intervals achieve coverage similar to those of Schmertmann et al., the baseline method whose intervals come closest to nominal coverage. Consistent with Bohk-Ewald et al. (2018), I find that most of the compared methods produce intervals that are far too narrow. The intervals of Myrskylä et al. and Ševčíková et al. capture only a minority of forecasted points despite a nominal coverage of 95%, and their coverage deteriorates further as the forecast horizon lengthens, averaging about 20%. Schmertmann et al. and CCF-Net are also too narrow, but they hold steady at roughly 80% coverage across the forecast horizon. Lee falls between these two groups, with coverage hovering around 60%. Because Bohk-Ewald et al. (2018) do not report uncertainty intervals for the de Beer and freeze rates methods, I do not produce intervals for them here.

These results show that CCF-Net not only produces competitive point forecasts, but also yields uncertainty intervals whose coverage matches that of the best-performing baseline. While its intervals are comparable to Schmertmann et al. (2014), they are still too narrow.

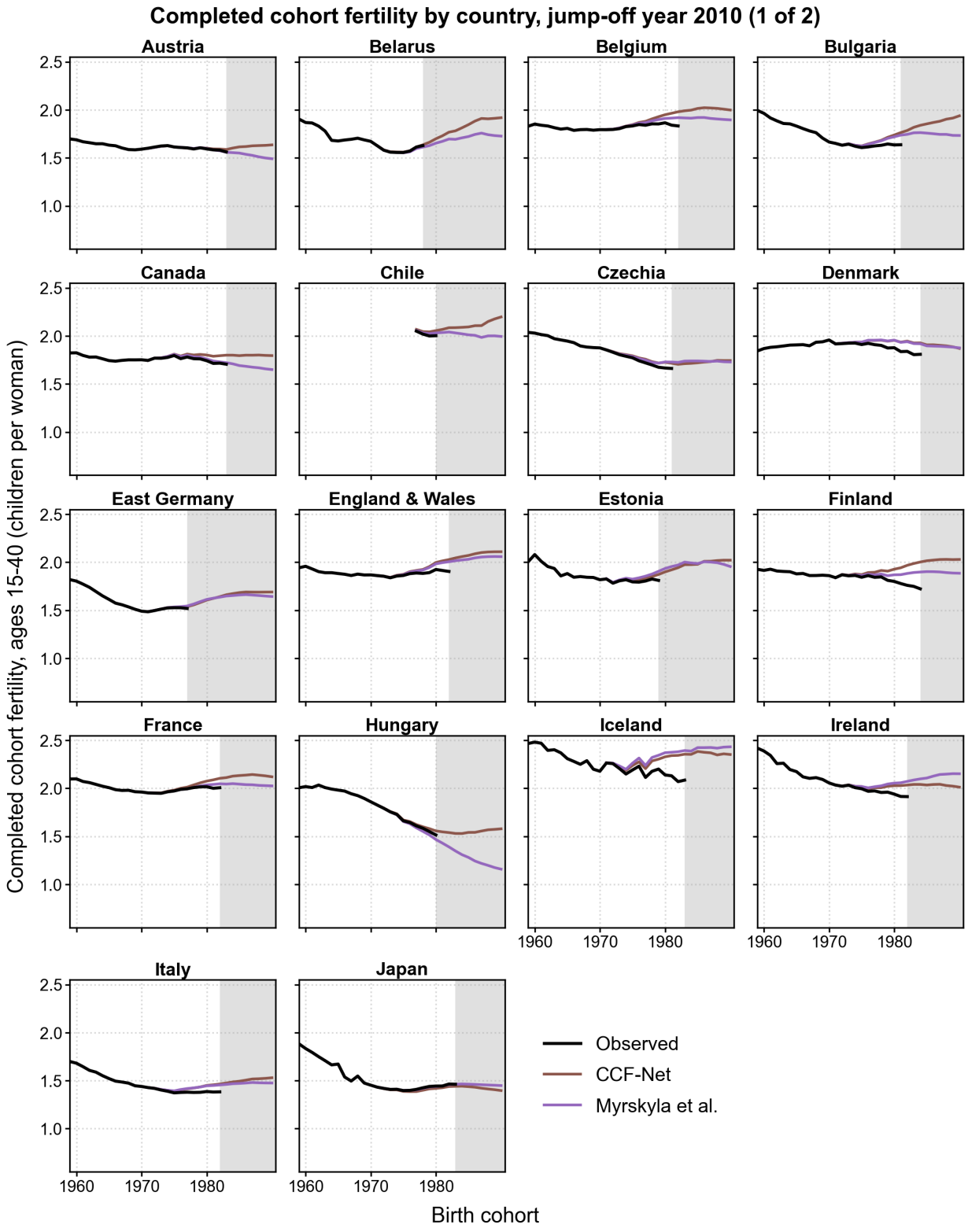



Figure A2: Completed cohort fertility by country from jump-off year 2010

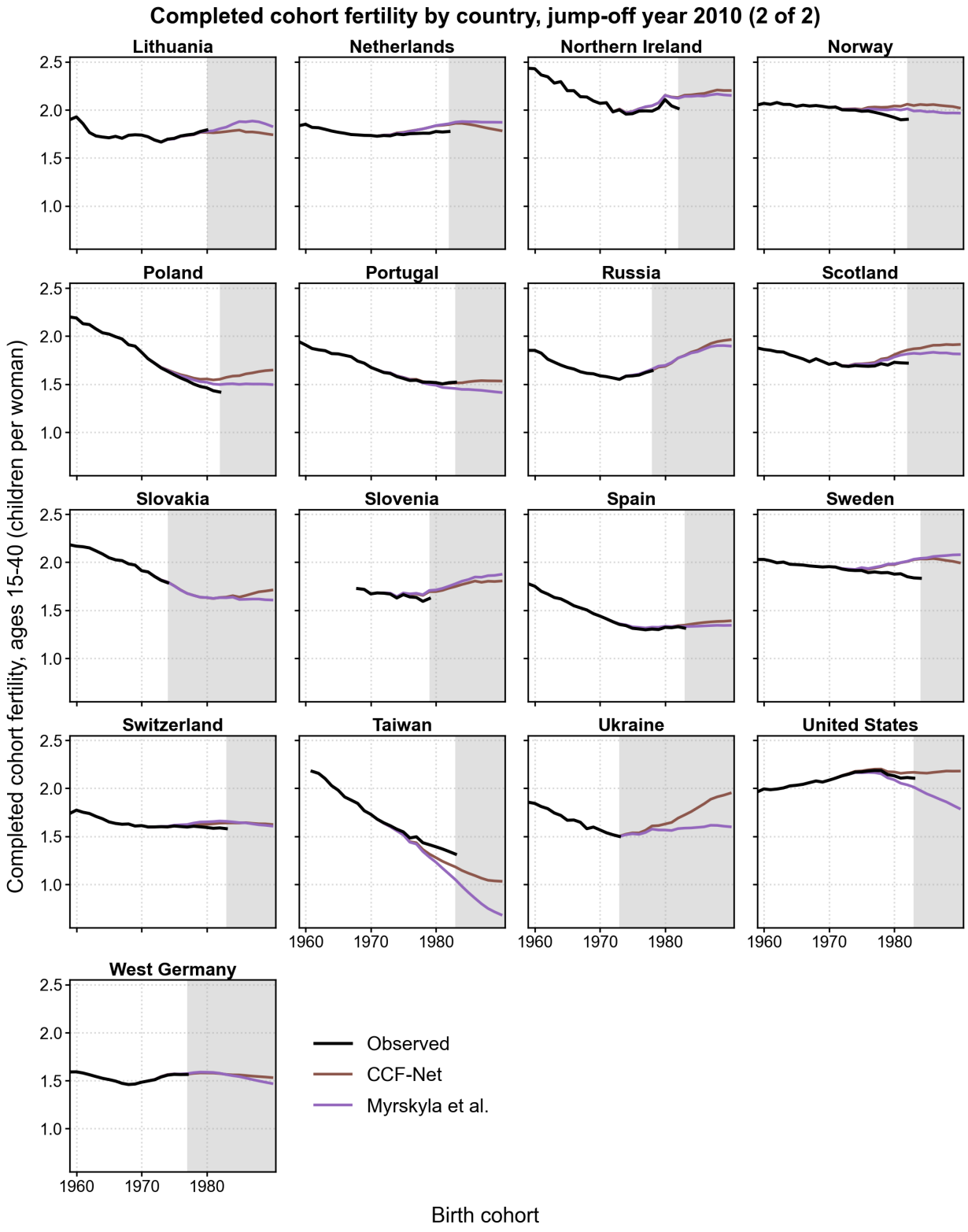



Figure A2: Completed cohort fertility by country from jump-off year 2010 (continued)